

Maximizing $K_r + I_r$ in graphs with fixed edge density

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Abstract

For every integer $r \geq 4$, and $\rho \in [0, 1]$, we asymptotically determine the maximum proportion of r -element sets of vertices that induce either a clique or an independent set in a large graph with density ρ . This generalizes a result of Olpp for $r = 3$.

After the initial idea for the main proof was found by the authors, various AI models were used to streamline the argument and perform the calculations necessary for completion of the proof.

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1 Introduction

For a graph G , write $v(G) = |V(G)|$, $e(G) = |E(G)|$ and $\rho(G) = e(G)/\binom{v(G)}{2}$ for its edge density. For a graph F , let $F(G)$ be the number of $v(F)$ -subsets of $V(G)$ that induce a copy of F , and write

$$t_{\text{ind}}(F, G) = \frac{F(G)}{\binom{v(G)}{v(F)}}$$

for the induced density of F in G . Denote by K_r the complete graph on r vertices and I_r the independent set on r vertices.

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A classical problem in extremal graph theory asks for the maximum (or minimum) proportion of copies of a graph (or a linear combination of some collection of graphs) in a large graph G with given edge density. See [1, 2, 4, 5, 6, 11, 13, 14, 9] for some papers on these topics. For example, Goodman [7] determined the minimum of $K_3(G) + I_3(G)$, and he asked for the maximum of $K_3(G) + I_3(G)$ among graphs G with a prescribed number of edges. Olpp [12] determined this maximum, showing that, up to the adjustments needed to obtain the prescribed edge count, it is attained by the disjoint union of a clique and isolated vertices or by its complement. We prove the analogous asymptotic result for $K_r(G) + I_r(G)$ for every fixed $r \geq 3$.

Let $L = L(n, \gamma)$ be the n -vertex graph consisting of a clique on $\lfloor \gamma n \rfloor$ vertices and $n - \lfloor \gamma n \rfloor$ isolated vertices, and let $\bar{L} = \bar{L}(n, \gamma)$ be its complement; see Figure 1. The edge density of L is $\gamma^2 + O(n^{-1})$, and of $\bar{L}$ is $1 - \gamma^2 + O(n^{-1})$.

Moreover, $K_r(\bar{L}) + I_r(\bar{L}) = K_r(L) + I_r(L)$ equals

$$\binom{\lfloor \gamma n \rfloor}{r} + \binom{n - \lfloor \gamma n \rfloor}{r} + \lfloor \gamma n \rfloor \binom{n - \lfloor \gamma n \rfloor}{r-1} \geq (\gamma^r + (1 - \gamma)^r + r\gamma(1 - \gamma)^{r-1}) \binom{n}{r} - O_r(n^{r-1}).$$

We show that the asymptotic maximum of $K_r(G) + I_r(G)$ over all n -vertex graphs G with edge density $\rho + o(1)$ is attained by $L(n, \sqrt{\rho})$ when $\rho > 1/2$ and by $\bar{L}(n, \sqrt{1 - \rho})$ when $\rho \leq 1/2$; see Figure 2. All asymptotic notation in the theorem statement below is taken as $n \rightarrow \infty$.

Theorem 1.1. *Fix an integer $r \geq 3$ and a real $\rho \in [0, 1]$. If G_n is an n -vertex graph with edge density $\rho + o(1)$, then it holds that*

$$\frac{K_r(G_n) + I_r(G_n)}{\binom{n}{r}} \leq \gamma^r + (1 - \gamma)^r + r\gamma(1 - \gamma)^{r-1} + o(1), \quad \text{where } \gamma = \begin{cases} \sqrt{1 - \rho} & \text{for } \rho \leq \frac{1}{2}, \text{ and} \\ \sqrt{\rho} & \text{otherwise.} \end{cases}$$

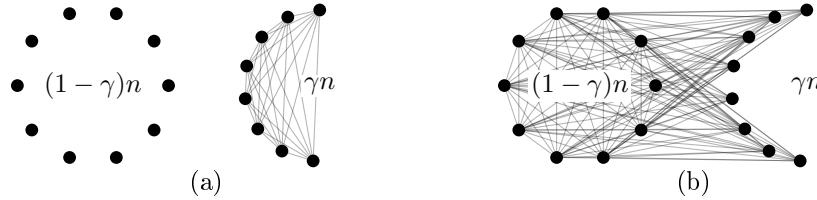


Figure 1: The extremal constructions: (a) the disjoint union of a clique of size approximately γn and $(1 - \gamma)n$ isolated vertices; (b) the complement of (a).

Generative AI statement: The extremal example in Theorem 1.1 has a two-part structure. Before using generative AI, the authors had a proof through the reduction to a three-part structure, and the remaining step was to solve a constrained optimization problem involving a two-variable polynomial of degree r (see Lemma 3.2 for the program). Models including Gemini Pro and ChatGPT supplied initial optimization arguments for completing this optimization and hence the three-part case; some suggestions were incorrect and were discarded. The final manuscript uses AI-originated optimization arguments both for this final step and for a new proof of the reduction from many parts to three. Through iteration with the authors, Prism, using GPT-5.6 Sol, helped refine the valid arguments and exposition. Some passages originated as AI-generated text, and much of the manuscript received lighter AI-assisted editing. The authors independently verified and take full responsibility for all arguments and text.

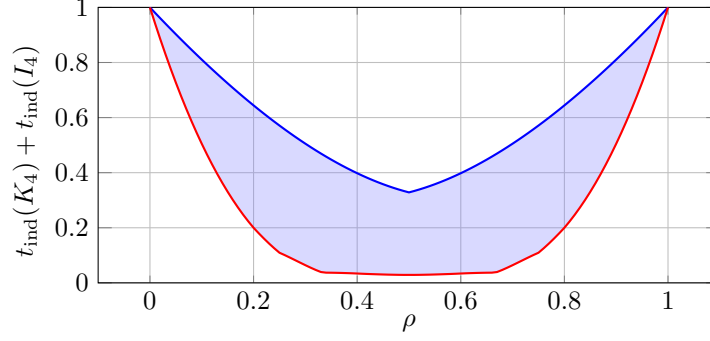


Figure 2: The extremal profile of $t_{\text{ind}}(K_4) + t_{\text{ind}}(I_4)$. The blue curve is the asymptotic upper envelope proved in this paper; the red curve shows values suggested by numerical experiments for the lower envelope, whose exact value is unknown.

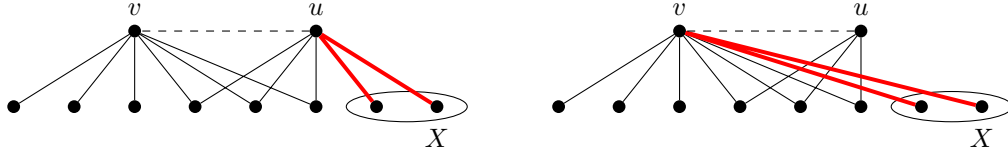


Figure 3: The Kelmans transformation from u to v .

1.1 Tools

Let G be a graph, and let u and v be distinct vertices of G . The *Kelmans transformation of G from u to v* is the graph G_{uv} obtained by deleting ux and adding vx for every

$$x \in N(u) \setminus (N(v) \cup \{v\});$$

see [8] and Figure 3. Note that for the transformation it does not matter if uv was an edge or not.

Csikvari proved the following monotonicity result.

Theorem 1.2 (Csikvari [3], Theorem 5.10). *For every pair of distinct vertices $u, v \in V(G)$ and every integer $r \geq 1$, the Kelmans transformation G_{uv} of G from u to v satisfies*

$$K_r(G_{uv}) \geq K_r(G) \quad \text{and} \quad I_r(G_{uv}) \geq I_r(G).$$

A *threshold graph* is a graph for which a vertex ordering $v_1, \dots, v_n$ with $d(v_1) \geq d(v_2) \geq \dots \geq d(v_n)$ satisfies

$$N(v_j) \setminus \{v_i\} \subseteq N(v_i) \setminus \{v_j\}$$

for every $1 \leq i < j \leq n$.

Theorem 1.3 (Csikvari [3], Theorem 2.6(a)). *Every graph can be transformed into a threshold graph by a sequence of Kelmans transformations.*

Since each Kelmans transformation preserves the number of edges, Theorems 1.2 and 1.3 give the following.

Corollary 1.4. *For all integers r, m, n with $r, n \geq 1$ and $0 \leq m \leq \binom{n}{2}$, there is a threshold graph G on n vertices with m edges that maximizes $K_r(G) + I_r(G)$ among all such graphs.*

We use graphons and step graphons in our proofs; for further background and the fundamental results of graph limit theory, see [10]. A *graphon* is a symmetric measurable function $W: [0, 1]^2 \rightarrow [0, 1]$. For $r \geq 1$, we write $t_{\text{ind}}(K_r, W)$ and $t_{\text{ind}}(I_r, W)$ for its induced clique and independent-set densities, respectively. Set

$$\Psi_r(W) = t_{\text{ind}}(K_r, W) + t_{\text{ind}}(I_r, W), \quad \rho(W) = \int_{[0,1]^2} W(x, y) \, dx \, dy.$$

Here $\rho(W)$ is the edge density of W . A graphon W is called a *step graphon* if there is a finite measurable partition $[0, 1] = P_1 \cup \dots \cup P_s$ such that W is constant on every $P_i \times P_j$. The sets P_i are called its *parts*. A 0–1 step graphon is a step graphon whose block values all belong to $\{0, 1\}$. For a graph G on the ordered vertex set $\{1, \dots, n\}$, its associated step graphon is obtained by taking n equal consecutive parts and assigning the value 1 to $P_i \times P_j$ exactly when $ij \in E(G)$; its diagonal blocks are assigned the value 0. We call W *nested* if it has a representative satisfying

$$W(x, z) \geq W(y, z) \quad \text{for almost every } (x, y, z) \text{ with } x \leq y.$$

For a nested step graphon, we always use the ordered representative in which the sets of points having identical rows are consecutive intervals. By symmetry, every row of a nested 0–1 step graphon is then the indicator of an initial interval, up to a null set. We call the resulting symmetric monotone 0–1 block pattern its *staircase pattern*.

2 Proof of Theorem 1.1

Fix an integer $r \geq 3$, and let

$$g_r(t) := t^r + (1-t)^r + rt(1-t)^{r-1} \quad \text{and} \quad M_r(\rho) := g_r\left(\sqrt{\max\{\rho, 1-\rho\}}\right).$$

Suppose, for the sake of contradiction, that there are $\eta > 0$ and, for arbitrarily large integers k , k -vertex graphs G_k such that

$$\frac{K_r(G_k) + I_r(G_k)}{\binom{k}{r}} > M_r(\sigma_k) + \eta, \quad \sigma_k = \rho(G_k) = \frac{e(G_k)}{\binom{k}{2}}.$$

By Corollary 1.4, we may assume that each G_k is a threshold graph and its vertices are ordered so that their degrees are non-increasing. For the associated step graphon, the ordering implies that its diagonal blocks can be set to 1 on an initial segment and to 0 thereafter so that every column is non-increasing. The resulting nested 0–1 step graphon V_k has at most k parts and differs from the associated graphon only on a set of area at most $1/k$. Put $\rho_k = \rho(V_k)$. The alteration of the diagonal blocks, together with the difference between sampling vertices with and without replacement, gives

$$\rho_k = \sigma_k + O(1/k), \quad \Psi_r(V_k) = \frac{K_r(G_k) + I_r(G_k)}{\binom{k}{r}} + O_r(1/k).$$

Since M_r is continuous on $[0, 1]$, it follows that, for all sufficiently large k under consideration,

$$\Psi_r(V_k) > M_r(\rho_k) + \frac{\eta}{2}.$$

For each such k , choose a nested 0–1 step graphon W_k with at most k parts and edge density ρ_k that maximizes Ψ_r in this finite-dimensional class. Such a maximizer exists because there are finitely many

staircase patterns and, allowing zero-length parts, each pattern has a compact set of admissible block lengths on which Ψ_r is continuous. Since V_k belongs to this class,

$$\Psi_r(W_k) \geq \Psi_r(V_k) > M_r(\rho_k) + \frac{\eta}{2}.$$

We will prove below that every such maximizer satisfies $\Psi_r(W_k) \leq M_r(\rho_k)$, which is the desired contradiction.

For $x \in [0, 1]$, let

$$d_k(x) = \int_0^1 W_k(x, y) dy$$

be its graphon degree. In the ordered nested representative, the neighbors of x form the initial interval $[0, d_k(x))$, up to a null set. Thus the measure of its neighbors preceding it is $\min\{x, d_k(x)\}$. Counting each edge by its later endpoint and recalling that graphon edge density counts ordered pairs gives

$$\rho_k = 2 \int_0^1 \min\{x, d_k(x)\} dx. \quad (1)$$

Similarly, an r -clique is counted by its last point and an independent r -set by its first point. Nestedness ensures that the other $r - 1$ points form a clique or an independent set, respectively. Therefore

$$t_{\text{ind}}(K_r, W_k) = r \int_0^1 \min\{x^{r-1}, d_k(x)^{r-1}\} dx$$

and

$$t_{\text{ind}}(I_r, W_k) = r \int_0^1 \min\{(1-x)^{r-1}, (1-d_k(x))^{r-1}\} dx.$$

Consequently, subject to (1), W_k maximizes

$$r \int_0^1 \min\{x^{r-1}, d_k(x)^{r-1}\} dx + r \int_0^1 \min\{(1-x)^{r-1}, (1-d_k(x))^{r-1}\} dx. \quad (2)$$

After merging consecutive intervals with identical rows, call the resulting intervals the canonical parts. Suppose that W_k has $s \geq 4$ such parts. If its last row is nonzero, then that row begins with a 1; symmetry and nestedness force the first row to be identically 1. Thus either W_k already has a zero last row or its complement followed by order reversal does. Let $\widehat{W}_k$ be the resulting graphon, and let $\widehat{\rho}_k$ be its edge density. Then

$$\widehat{\rho}_k \in \{\rho_k, 1 - \rho_k\}, \quad \Psi_r(\widehat{W}_k) = \Psi_r(W_k).$$

Moreover, $\widehat{W}_k$ is a maximizer in the corresponding class at density $\widehat{\rho}_k$, and formulas (1)–(2) apply with W_k, ρ_k replaced by $\widehat{W}_k, \widehat{\rho}_k$.

For $i \in \{1, \dots, s\}$, let t_i be the number of canonical parts on which the i th row of $\widehat{W}_k$ equals 1. Nestedness and distinctness give

$$t_1 > t_2 > \dots > t_s = 0.$$

Since the last row is zero, symmetry makes the first row zero on the last part, so $t_1 \leq s - 1$. Hence necessarily $t_i = s - i$ for every i . In particular, the first column is 1 except on the last part, and the second column is 1 except on the last two parts. Let a, b be the lengths of the first two parts and c, d the lengths of the last two. These lengths are positive and $a + b + c + d \leq 1$.

With these parameters, the four forced outer blocks are

$$\widehat{W}_k(x, y) = \begin{cases} 1 & \text{if } x \in [0, 1 - d) \text{ and } y \in [0, a), \\ 0 & \text{if } x \in [1 - d, 1] \text{ and } y \in [0, a), \\ 1 & \text{if } x \in [0, 1 - c - d) \text{ and } y \in [a, a + b), \\ 0 & \text{if } x \in [1 - c - d, 1] \text{ and } y \in [a, a + b). \end{cases}$$

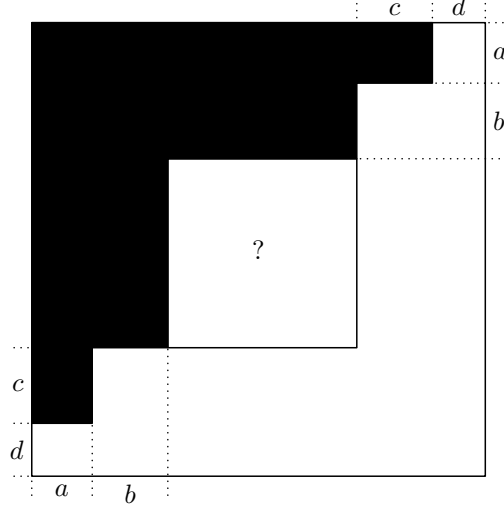


Figure 4: The forced corner blocks of the ordered nested graphon $\widehat{W}_k$.

The contribution of the vertices $x \in [0, a+b] \cup [1-c-d, 1]$ to the right-hand side of (1) is $(a+b)^2 + 2ac$, while their contribution to (2) is

$$(a+b)^r + rca^{r-1} + rad^{r-1} + rb(c+d)^{r-1} + (c+d)^r. \quad (3)$$

Let

$$f(w, x, y, z) := (w+x)^r + ryw^{r-1} + r wz^{r-1} + rx(y+z)^{r-1} + (y+z)^r.$$

For any non-negative quadruple (w, x, y, z) satisfying

$$w+x = a+b, \quad y+z = c+d, \quad (w+x)^2 + 2wy = (a+b)^2 + 2ac,$$

retain the same staircase pattern and all intervening parts, but give the first two and last two parts lengths w, x, y, z . The first two equalities keep the total outer length, and hence all intervening parts, fixed. The third equality preserves the outer contribution to the edge density. All other terms in (2) are unchanged, while equation (3) shows that the varying part is exactly f . Thus every feasible quadruple is realized by an admissible nested step graphon with the same total edge density.

Because $\widehat{W}_k$ maximizes (2) subject to (1), the tuple (a, b, c, d) maximizes f over all such quadruples.

Put

$$p = a+b, \quad q = c+d, \quad \alpha = p^2 + 2ac.$$

Since $a, b, c, d > 0$ and $p+q \leq 1$, we have $p, q \in (0, 1)$. Moreover, $a < p$ and, because $d > 0$, $c < q \leq 1-p$. Therefore

$$p^2 < \alpha < p^2 + 2p(1-p) = 1 - (1-p)^2 < 1.$$

Thus all the domain assumptions of Lemma 3.1 are satisfied.

Lemma 3.1 now implies that at least one coordinate of the maximizer (a, b, c, d) is zero, contradicting $a, b, c, d > 0$.

Therefore, every W_k has at most three canonical parts. Each W_k has one of the two structures depicted in Figure 5, with parameters $p, q \geq 0$ satisfying $p+q \leq 1$.

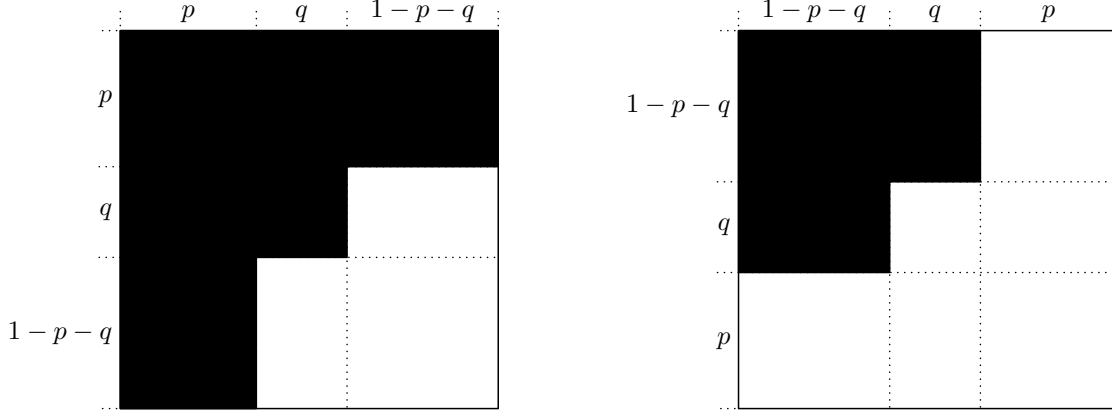


Figure 5: The possible structures of a maximizing graphon W_k .

For both structures we have

$$\Psi_r(W_k) = (p+q)^r + r(1-p-q)p^{r-1} + rq(1-p-q)^{r-1} + (1-p-q)^r,$$

and

$$e = (p+q)^2 + 2p(1-p-q)$$

is either ρ_k or $1 - \rho_k$.

If $e \in \{0, 1\}$, then W_k is respectively empty or complete up to complementation, and $\Psi_r(W_k) = 1 = M_r(\rho_k)$. Suppose that $0 < e < 1$. Then Lemma 3.2 gives

$$\Psi_r(W_k) \leq \max_{\gamma \in \{\sqrt{e}, \sqrt{1-e}\}} g_r(\gamma).$$

The two possible values of γ have squares summing to 1. Lemma 3.4 therefore shows that their maximum is attained at $\gamma = \sqrt{\max\{e, 1-e\}}$. Since $\{e, 1-e\} = \{\rho_k, 1-\rho_k\}$, we obtain

$$\Psi_r(W_k) \leq M_r(\rho_k).$$

This contradicts the strict inequality $\Psi_r(W_k) > M_r(\rho_k) + \eta/2$ obtained above. Hence the uniform upper bound holds, completing the proof of Theorem 1.1.

3 Auxiliary optimization (AI-assisted proofs)

The proofs in this section contain the optimization arguments. Section 4 contains the differentiations and algebraic identities used here, separating those routine computations from the sign, compactness, and endpoint arguments.

Lemma 3.1. *Fix an integer $r \geq 3$ and three positive reals p, q and α that satisfy the conditions $p+q \leq 1$ and $p^2 < \alpha < 1 - (1-p)^2$, and let*

$$f(w, x, y, z) := (w+x)^r + ryw^{r-1} + rwz^{r-1} + rx(y+z)^{r-1} + (y+z)^r.$$

If (w^*, x^*, y^*, z^*) is a global maximizer of the program

$$(P) = \begin{cases} \text{maximize} & f(w, x, y, z) \\ \text{subject to} & w + x = p \\ & y + z = q \\ & (w + x)^2 + 2wy = \alpha \\ & (w, x, y, z) \geq 0, \end{cases}$$

then at least one of w^* , x^* , y^* , and z^* is equal to zero.

Proof. Let $C = (\alpha - p^2)/2$. On the constraint set we have

$$x = p - w, \quad z = q - y, \quad C = (\alpha - p^2)/2 = ((w + x)^2 + 2wy - (w + x)^2)/2 = wy.$$

Suppose for contradiction that all four coordinates of a maximizer are positive. Then $0 < C = wy < pq$, and

$$(w, x, y, z) = \left(w, p - w, \frac{C}{w}, q - \frac{C}{w} \right), \quad \frac{C}{q} < w < p,$$

since the values $w = C/q$ and $w = p$ correspond to $z = 0$ and $x = 0$, respectively. On this curve, the objective is to maximize

$$f\left(w, p - w, \frac{C}{w}, q - \frac{C}{w}\right) = p^r + r \cdot \frac{C}{w} \cdot w^{r-1} + rw \left(q - \frac{C}{w}\right)^{r-1} + r(p - w)q^{r-1} + q^r = p^r + q^r + rpq^{r-1} + rh(w),$$

where

$$h(w) = Cw^{r-2} + w \left(q - \frac{C}{w}\right)^{r-1} - wq^{r-1}.$$

Thus, maximizing the objective is equivalent to maximizing $h(w)$ on the open interval $p > w > C/q$. However, Lemma 4.1 shows that h is strictly convex in this interval, so it cannot attain its maximum there; a contradiction. $\square$

Lemma 3.2. Fix an integer $r \geq 3$ and a real $e \in (0, 1)$, and let

$$F(p, q) := (p + q)^r + r(1 - p - q)p^{r-1} + rq(1 - p - q)^{r-1} + (1 - p - q)^r.$$

If (p^*, q^*) is a global maximizer of the program

$$(P) = \begin{cases} \text{maximize} & F(p, q) \\ \text{subject to} & p + q \leq 1 \\ & (p + q)^2 + 2p(1 - p - q) = e \\ & (p, q) \geq 0, \end{cases}$$

then $(p^*, q^*) \in \{(0, \sqrt{e}), (1 - \sqrt{1 - e}, 0)\}$.

Proof. Put $y = 1 - p - q$. Since $e < 1$, every feasible point has $y > 0$ (note that $y = 0$ yields $e = 1$). The constraints $e = (1 - y)^2 + 2py$ and $p + q = 1 - y$ yield the following re-parametrization

$$p(y) = \frac{e - (1 - y)^2}{2y}, \quad q(y) = 1 - p - y = \frac{2y - 2py - 2y^2}{2y} = \frac{1 - (1 - y)^2 - 2py - y^2}{2y} = \frac{1 - e - y^2}{2y}.$$

The conditions $p, q \geq 0$ are equivalent to

$$1 - \sqrt{e} \leq y \leq \sqrt{1 - e}.$$

Thus the feasible curve $(p(y), q(y))$ is a continuous image of a closed interval and is therefore compact. Its endpoints are

$$(p, q) = (0, \sqrt{e}) \quad \text{and} \quad (p, q) = (1 - \sqrt{1 - e}, 0).$$

The function $F(p, q)$ is continuous, so it attains a maximum on the feasible set. To locate that maximum, define

$$\Phi(y) = F(p(y), q(y)) = (1 - y)^r + ry \cdot p(y)^{r-1} + rq(y) \cdot y^{r-1} + y^r.$$

Thus maximizing (P) is equivalent to maximizing $\Phi(y)$ on the interval $[1 - \sqrt{e}, \sqrt{1 - e}]$. Since $\Phi''(y) > 0$ for all y such that $\Phi'(y) = 0$ by Lemma 4.2, every critical point of Φ inside the interval is a strict local minimum. Therefore, a global maximum of Φ occurs at one of the endpoints. $\square$

Lemma 3.3. *Fix an integer $r \geq 3$, and let*

$$f(t) := t^{r-2} - (r-1)(1-t)^{r-2}, \quad t_0 := \frac{(r-1)^{1/(r-2)}}{1 + (r-1)^{1/(r-2)}} \quad \text{and} \quad D(s) := f\left(\sqrt{1-s^2}\right) + f(s).$$

Then $D(0) < 0$, $D(t_0) > 0$ and $D : [0, t_0] \rightarrow \mathbb{R}$ has exactly one zero.

Proof. We have $D(0) = f(1) + f(0) = 1 - (r-1) = 2 - r < 0$. Lemma 4.3 shows that f is strictly increasing on $[0, 1]$, $f(t_0) = 0$, and $t_0 < 1/\sqrt{2}$. Therefore $\sqrt{1-t_0^2} > t_0$, and

$$D(t_0) = f\left(\sqrt{1-t_0^2}\right) + f(t_0) = f\left(\sqrt{1-t_0^2}\right) > f(t_0) = 0.$$

Let

$$Z = \{u \in [0, t_0] : D(u) = 0\}.$$

Since D is a continuous map on $[0, t_0]$, the opposite signs at the endpoints and the intermediate value theorem show that Z is nonempty. Moreover, every $u \in Z$ satisfies the assumptions of Lemma 4.4 and thus $D'(u) > 0$. Finally, the set $D^{-1}(\{0\})$ is a closed subset of $[0, t_0]$, and hence $\inf Z \in Z$. Let $u_1 = \min Z$. The derivative at u_1 shows that D is positive on $(u_1, u_1 + \delta)$ for some $0 < \delta < t_0 - u_1$. If $Z \setminus \{u_1\} \neq \emptyset$, then the nonempty closed set $Z \cap [u_1 + \delta, t_0]$ would have a minimum u_2 . There is no zero of D in (u_1, u_2) , so the intermediate value theorem yields that $D > 0$ throughout this interval. However, $D'(u_2) > 0$ yields that D must be negative slightly to the left of u_2 ; a contradiction. We conclude that u_1 is the unique zero of D . $\square$

Lemma 3.4. *Fix an integer $r \geq 3$, and let*

$$g_r(t) := t^r + (1-t)^r + rt(1-t)^{r-1}.$$

If x and y are non-negative reals satisfying $x \geq y$ and $x^2 + y^2 = 1$, then $g_r(x) \geq g_r(y)$.

Proof. Let

$$f(t) := t^{r-2} - (r-1)(1-t)^{r-2}.$$

Lemma 4.3 states that f has a unique zero t_0 where $t_0 < 1/\sqrt{2}$, and that $g_r(t)$ decreases when $t < t_0$ and increases when $t > t_0$. The assumption $x^2 + y^2 = 1$ implies

$$0 \leq y \leq \frac{1}{\sqrt{2}} \leq x \leq 1.$$

If $y \geq t_0$, then as both x, y lie in the increasing region of g_r and $x \geq y$, we have $g_r(x) \geq g_r(y)$.

It remains to handle the case $y < t_0$. For $s \in [0, t_0]$, set

$$H(s) = g_r\left(\sqrt{1-s^2}\right) - g_r(s), \quad D(s) = f\left(\sqrt{1-s^2}\right) + f(s).$$

Our aim is to prove that $H(s) \geq 0$ for $s \in [0, t_0]$. Using $g'_r(t) = rtf(t)$ from Lemma 4.3, the chain rule gives

$$H'(s) = -\frac{s}{\sqrt{1-s^2}} \cdot g'_r\left(\sqrt{1-s^2}\right) - g'_r(s) = -rsf\left(\sqrt{1-s^2}\right) - rsf(s) = -rsD(s).$$

Thus, for all $s \in (0, t_0]$, the derivative $H'(s)$ has the opposite sign from $D(s)$. Let s_0 be the unique zero of D given by Lemma 3.3. The conclusion of Lemma 3.3 translates to

$$H'(s) > 0 \quad \forall s \in (0, s_0), \quad \text{and} \quad H'(s) < 0 \quad \forall s \in (s_0, t_0).$$

Therefore, H is strictly increasing on $[0, s_0]$ and strictly decreasing on $[s_0, t_0]$. Consequently, the minimum of H on $[0, t_0]$ is attained at one of the endpoints of the interval. Clearly, $H(0) = g_r(1) - g_r(0) = 0$. On the other hand, we claim that $H(t_0) > 0$. Indeed, since $\sqrt{1-t_0^2} > t_0$ and $g_r(x)$ is strictly increasing for $x \geq t_0$, it holds that

$$H(t_0) = g_r\left(\sqrt{1-t_0^2}\right) - g_r(t_0) > 0.$$

We conclude that $H(s) \geq 0$ on $[0, t_0]$, which finishes the proof. $\square$

4 Routine computations (AI-assisted proofs)

In this section we present the computational parts used in our proofs.

Lemma 4.1. *Fix an integer $r \geq 3$ and two positive reals q and C . Let*

$$h(w) := Cw^{r-2} + w\left(q - \frac{C}{w}\right)^{r-1} - wq^{r-1}.$$

If $w > C/q$ then $h''(w) > 0$.

Proof. The product and chain rules give

$$h'(w) = (r-2)Cw^{r-3} + \left(q - \frac{C}{w}\right)^{r-1} + (r-1)\frac{C}{w}\left(q - \frac{C}{w}\right)^{r-2} - q^{r-1}.$$

Differentiating the second term gives

$$(r-1)\frac{C}{w^2}\left(q - \frac{C}{w}\right)^{r-2},$$

whereas differentiating the third term gives

$$-(r-1)\frac{C}{w^2}\left(q - \frac{C}{w}\right)^{r-2} + (r-1)(r-2)\frac{C^2}{w^3}\left(q - \frac{C}{w}\right)^{r-3}.$$

The first term here cancels the term obtained above; thus,

$$h''(w) = (r-2)(r-3)C \cdot w^{r-4} + (r-1)(r-2) \cdot \frac{C^2}{w^3} \cdot \left(q - \frac{C}{w}\right)^{r-3}.$$

The first summand is non-negative for all $r \geq 3$, and thus it remains to show that the second summand is strictly positive. Since

$$\frac{w \left(q - \frac{C}{w} \right)}{q} = w - \frac{C}{q} > 0,$$

we conclude that the term $\left(q - \frac{C}{w} \right)$ is positive, which together with $w > 0$, $C > 0$ and $r > 2$ concludes the proof. $\square$

Lemma 4.2. *Fix an integer $r \geq 3$ and a real number $e \in (0, 1)$. For $y \in (1 - \sqrt{e}, \sqrt{1 - e})$, let*

$$p = p(y) := \frac{e - (1 - y)^2}{2y}, \quad q = q(y) := \frac{1 - e - y^2}{2y}, \quad x = x(y) := 1 - y = p + q,$$

and

$$\Phi(y) := x^r + ryp^{r-1} + rqq^{r-1} + y^r.$$

If $\Phi'(y) = 0$, then $\Phi''(y) > 0$.

Proof. Inside the open interval $(1 - \sqrt{e}, \sqrt{1 - e})$, we have $p, q, y > 0$, and

$$p' = p'(y) = \frac{q}{y}, \quad q' = q'(y) = \frac{p-1}{y}, \quad x' = x'(y) = -1.$$

Indeed, rewriting p and q gives

$$p = \frac{e-1}{2y} + 1 - \frac{y}{2}, \quad q = \frac{1-e}{2y} - \frac{y}{2}.$$

Differentiating then gives

$$p' = -\frac{e-1}{2y^2} - \frac{1}{2} = \frac{q}{y}, \quad q' = -\frac{1-e}{2y^2} - \frac{1}{2} = \frac{p-1}{y}.$$

Differentiating the four summands of Φ separately gives

$$\Phi'(y) = r \left[x^{r-1}x' + p^{r-1} + (r-1)yp^{r-2}p' + q'y^{r-1} + (r-1)qq^{r-2} + y^{r-1} \right].$$

Substituting $x' = -1$, $p' = q/y$, and $q' = (p-1)/y$ produces the first expression below

$$\begin{aligned} \Phi'(y) &= r \left[-x^{r-1} + p^{r-1} + (r-1)qp^{r-2} + (p-1)y^{r-2} + (r-1)q \cdot y^{r-2} + y^{r-1} \right] \\ &= r \left[p^{r-1} + (r-1)qp^{r-2} + (r-2)qy^{r-2} - x^{r-1} \right], \end{aligned} \tag{4}$$

and for the second equality we used $p + q - 1 = -y$.

At a critical point of Φ , i.e., when $\Phi'(y) = 0$, substituting $x = p + q$ into (4) gives

$$(r-2)qy^{r-2} = (p+q)^{r-1} - p^{r-1} - (r-1)qp^{r-2}.$$

Expanding $(p+q)^{r-1}$ and dividing by $q > 0$ yield that when $\Phi'(y) = 0$, then

$$(r-2)y^{r-2} = \sum_{k=2}^{r-1} \binom{r-1}{k} p^{r-1-k} q^{k-1}. \tag{5}$$

Now we compute the second derivative of $\Phi(y)$:

$$\begin{aligned}\Phi''(y) &= r(r-1)q \cdot \frac{p^{r-2}}{y} + r(r-1)(p-1)\frac{p^{r-2}}{y} + r(r-1)(r-2)p^{r-3} \cdot \frac{q^2}{y} \\ &\quad + r(r-2)(p-1)y^{r-3} + r(r-2)^2qy^{r-3} + r(r-1)x^{r-2}.\end{aligned}$$

Using the identity $p+q-1=-y$ twice in the following reordering

$$\begin{aligned}\Phi''(y) &= r(r-1)(q+p-1) \cdot \frac{p^{r-2}}{y} + r(r-1)(r-2)p^{r-3} \cdot \frac{q^2}{y} \\ &\quad + r(r-2)(p-1+q)y^{r-3} + r(r-2)(r-3)qy^{r-3} + r(r-1)x^{r-2},\end{aligned}$$

we obtain

$$\Phi''(y) = r(r-1)(x^{r-2} - p^{r-2}) + r(r-1)(r-2)p^{r-3} \cdot \frac{q^2}{y} + r(r-2)(r-3)qy^{r-3} - r(r-2)y^{r-2}. \quad (6)$$

First, note that the second summand of Φ'' is positive and the third summand is non-negative. Next, using $x = p + q$, the first summand of Φ'' can be rewritten as follows:

$$r(r-1)(x^{r-2} - p^{r-2}) = r(r-1) \sum_{j=1}^{r-2} \binom{r-2}{j} p^{r-2-j} q^j. \quad (7)$$

Finally, at every critical point of Φ , i.e., when $\Phi'(y) = 0$, the last summand of Φ'' can be rewritten using (5) in the following way:

$$-r(r-2)y^{r-2} = -r \sum_{j=1}^{r-2} \binom{r-1}{j+1} p^{r-2-j} q^j. \quad (8)$$

We claim that the sum of (7) and (8) is strictly positive. Indeed, using

$$(r-1) \binom{r-2}{j} = (j+1) \binom{r-1}{j+1}$$

gives that in (6) the coefficient in front of $p^{r-2-j}q^j$, for $j \in [r-2]$, is equal to

$$r(r-1) \binom{r-2}{j} - r \binom{r-1}{j+1} = r((j+1)-1) \binom{r-1}{j+1} = rj \binom{r-1}{j+1} > 0.$$

Therefore, we conclude that $\Phi''(y) > 0$ at every critical point of Φ , i.e., when $\Phi'(y) = 0$. □

Lemma 4.3. *Fix an integer $r \geq 3$, and let*

$$g_r(t) = t^r + (1-t)^r + rt(1-t)^{r-1} \quad \text{and} \quad f(t) = t^{r-2} - (r-1)(1-t)^{r-2}.$$

It holds that $g'_r(t) = rtf(t)$, f is strictly increasing on $[0, 1]$, and f has the unique zero

$$t_0 = \frac{(r-1)^{1/(r-2)}}{1 + (r-1)^{1/(r-2)}} < \frac{1}{\sqrt{2}}.$$

In particular, g_r is strictly decreasing on $[0, t_0]$ and strictly increasing on $[t_0, 1]$.

Proof. Direct computation of the derivatives provides

$$g'_r(t) = r \cdot t^{r-1} - r \cdot (1-t)^{r-1} + r \cdot (1-t)^{r-1} - r \cdot t \cdot (r-1)(1-t)^{r-2} = r \cdot t \cdot f(t), \quad \text{and}$$

$$f'(t) = (r-2) [t^{r-3} + (r-1)(1-t)^{r-3}].$$

Since $f(0) = 1-r < 0 < f(1) = 1$ and $f'(t) > 0$ for $t \in [0, 1]$, f has exactly one zero in $[0, 1]$. Solving

$$\left(\frac{t}{1-t} \right)^{r-2} = r-1$$

yields the displayed formula for t_0 . To prove the claimed upper bound $1/\sqrt{2}$, note that $r-1 \leq 2^{r-2}$ for every $r \geq 3$. Thus $(r-1)^{1/(r-2)} \leq 2$, and hence $t_0 \leq 2/3 < 1/\sqrt{2}$.

Finally, because $rt > 0$ for $t \in (0, 1]$, the identity $g'_r(t) = rt f(t)$ shows that g'_r has the same sign as f . Since f is strictly increasing and vanishes only at t_0 , we have

$$g'_r(t) < 0 \quad \text{when } 0 < t < t_0, \quad \text{and} \quad g'_r(t) > 0 \quad \text{when } t_0 < t \leq 1,$$

which gives the asserted monotonicity of g_r . □

Lemma 4.4. *Fix a positive integer n , and let*

$$f(t) = t^n - (n+1)(1-t)^n, \quad D(s) = f\left(\sqrt{1-s^2}\right) + f(s).$$

For every $u \in (0, 1/\sqrt{2})$ such that $D(u) = 0$, it holds that $D'(u) > 0$.

Proof. To simplify the presentation of the proof, we introduce some auxiliary notation. Let $v := \sqrt{1-u^2}$, and note that $0 < u < v < 1$. Also, let

$$A := v \cdot (u(1-u))^{n-1} + v^{n-1} \cdot (1-u)^n \quad \text{and} \quad B := u \cdot (v(1-v))^{n-1} + u^{n-1} \cdot (1-v)^n,$$

which will be used to simplify the expression $v \cdot f'(u) - u \cdot f'(v)$.

Claim 4.5. $A > B$.

Proof. Indeed, since $v > u$ and $1-u > 1-v$, it is enough to observe that $u(1-u) > v(1-v)$. The relation $u^2 + v^2 = 1$, with $0 < u < v < 1$ and $u+v \geq 1$ gives that $|u-0.5| < |v-0.5|$, which implies $u(1-u) > v(1-v)$. □

Next, since

$$\frac{d}{ds} \sqrt{1-s^2} = -\frac{s}{\sqrt{1-s^2}},$$

the chain rule gives

$$D'(s) = f'(s) - \frac{s}{\sqrt{1-s^2}} \cdot f'\left(\sqrt{1-s^2}\right).$$

Evaluating at $s = u$ and using $\sqrt{1-u^2} = v$ yields $D'(u) = f'(u) - \frac{u}{v} f'(v)$. Therefore, in our quest to prove $D'(u) > 0$ under the assumption $D(u) = f(u) + f(v) = 0$, it suffices to establish the following identity:

$$\frac{v}{n} \cdot D'(u) = \frac{v \cdot f'(u) - u \cdot f'(v)}{n} = \frac{A-B}{(1-u)^n + (1-v)^n} > 0, \quad (9)$$

where the inequality holds due to the fact $A > B$.

Observe that evaluating $f(t)$ for $t = u$ and $t = v$ and rearranging yields that

$$f(u) + f(v) = 0 \iff n + 1 = \frac{u^n + v^n}{(1 - u)^n + (1 - v)^n}.$$

Moreover, a direct computation of the derivative of f shows that

$$f'(t) = n \cdot (t^{n-1} + (n + 1)(1 - t)^{n-1}).$$

Therefore, a tedious yet straightforward calculation reveals that

$$\begin{aligned} \frac{v \cdot f'(u) - u \cdot f'(v)}{n} &= v(u^{n-1} + (n + 1)(1 - u)^{n-1}) - u(v^{n-1} + (n + 1)(1 - v)^{n-1}) \\ &= vu^{n-1} - uv^{n-1} + (n + 1)(v(1 - u)^{n-1} - u(1 - v)^{n-1}) \\ &= \frac{(vu^{n-1} - uv^{n-1}) \cdot ((1 - u)^n + (1 - v)^n) + (u^n + v^n) \cdot (v(1 - u)^{n-1} - u(1 - v)^{n-1})}{(1 - u)^n + (1 - v)^n}. \end{aligned}$$

Therefore, our goal of establishing (9) reduces to showing the following two identities for A and B :

$$\begin{aligned} A &= (vu^{n-1} - uv^{n-1}) \cdot (1 - u)^n + (u^n + v^n) \cdot v(1 - u)^{n-1} \quad \text{and} \\ -B &= (vu^{n-1} - uv^{n-1}) \cdot (1 - v)^n - (u^n + v^n) \cdot u(1 - v)^{n-1}. \end{aligned}$$

Let us first verify the identity for A . Plugging in the definition of $A = v \cdot (u(1 - u))^{n-1} + v^{n-1} \cdot (1 - u)^n$, and then dividing both sides of the identity in question by the factor $v(1 - u)^{n-1}$, brings us to check the following (equivalent) identity:

$$u^{n-1} + (1 - u)v^{n-2} = (u^{n-1} - uv^{n-2}) \cdot (1 - u) + u^n + v^n = u^{n-1} - uv^{n-2} + (u^2 + v^2) \cdot v^{n-2}. \quad (10)$$

However, (10) readily follows from $u^2 + v^2 = 1$.

Similarly, we verify the identity for $-B$. Recalling that $B = u \cdot (v(1 - v))^{n-1} + u^{n-1} \cdot (1 - v)^n$, and then dividing both sides by the factor $-u(1 - v)^{n-1}$, brings us to the identity

$$v^{n-1} + (1 - v)u^{n-2} = u^n + v^n + (v^{n-1} - vu^{n-2}) \cdot (1 - v) = v^{n-1} - vu^{n-2} + (u^2 + v^2) \cdot u^{n-2}. \quad (11)$$

As in the case of (10), the identity (11) now follows from $u^2 + v^2 = 1$, which finishes the proof. $\square$

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